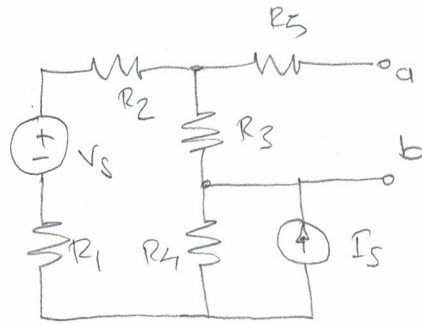


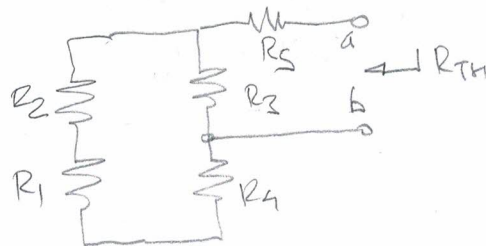
3.64)

SOIK THEORENÄQUIVALENZ



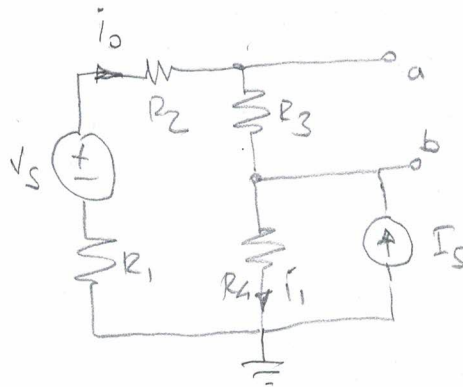
$$\begin{aligned} R_1 &= 10 \Omega \\ R_2 &= 8 \Omega \\ R_3 &= 5 \Omega \\ R_4 &= 4 \Omega \\ V_S &= 10 \text{ V} \\ I_S &= 2 \text{ A} \\ R_5 &= 1 \Omega \end{aligned}$$

Lösung: Zuerst R_{TH} , dann SPANNUNGSKITZ, DA BORT STRÖMKITZ



$$R_{TH} = R_5 + \frac{(R_1 + R_2 + R_4)R_3}{R_1 + R_2 + R_4 + R_3} \approx 5.074 \Omega$$

SPANNUNGEN V_{TH} ?



NODENANALYSE

$$a) \quad i_0 - i_2 = 0$$

$$b) \quad i_0 + I_S - i_1 = 0 \quad (1)$$

UTDRUCK I SPÄNNINGAR

$$i_1 = \frac{V_b}{R_4} \quad (2)$$

$$i_0 = \frac{V_a - V_b}{R_2} \quad (3)$$

KSPIL ZER

$$-R_1 i_0 + V_S - R_2 i_0 = V_a \quad (4)$$

(3) & (2) i (1)

$$\frac{V_a - V_b}{R_2} + I_S - \frac{V_b}{R_4} = 0 \quad (5)$$

(3) & (4)

$$-\frac{R_1 + R_2}{R_2} (V_a - V_b) + V_S = V_a \quad (6)$$

ELLER SOM

$$\begin{bmatrix} \frac{1}{R_3} & -\frac{1}{R_3} - \frac{1}{R_4} \\ -\frac{R_1 + R_2}{R_3} - 1 & \frac{R_1 + R_3}{R_3} \end{bmatrix} \begin{bmatrix} V_a \\ V_b \end{bmatrix} = \begin{bmatrix} -I_S \\ -V_S \end{bmatrix}$$

MED SIFFROR

$$\begin{bmatrix} \frac{1}{5} & -\frac{1}{5} - \frac{1}{4} \\ -\frac{18}{5} - 1 & \frac{18}{5} \end{bmatrix} \begin{bmatrix} V_a \\ V_b \end{bmatrix} = \begin{bmatrix} -2 \\ -10 \end{bmatrix}$$

LÖSNINGEN BLIR

$$\begin{bmatrix} V_a \\ V_b \end{bmatrix} = \begin{bmatrix} 8.667 \\ 8.296 \end{bmatrix} \text{ V}$$

$$V_{TH} = V_a - V_b = 8.667 - 8.296 = \underline{\underline{-0.371 \text{ V}}}$$

$$\underline{\text{SVAR:}} \quad V_{TH} = -0.371 \text{ V}$$

$$R_{TH} = 5.07 \text{ SL}$$